

Properties evolution and damage mechanism of SiC/SiC composites after thermal shock at 1300 °C

Yongjin Wei^a, Junwei Lan^a, Yuxi Yu^{a,*}, Liuying Huang^{a,**}, Dahai Ye^b, Zhongyi Fu^b

^a Fujian Key Laboratory of Advance Materials, Department of Materials Science and Engineering, College of Materials, Xiamen University, Xiamen, 361005, China

^b AEEC Hunan Aviation Powerplant Research Institute, Zhuzhou, 412002, China

ARTICLE INFO

Handling Editor: P. Vincenzini

Keywords:

Properties evolution
Damage mechanism
SiC/SiC composites
Thermal shock

ABSTRACT

SiC/SiC composites in aerospace applications are exposed to a variety of thermal conditions, in particular, thermal stress due to rapid heating and cooling. Therefore, it is crucial to investigate and evaluate the resistance of SiC/SiC composites to repeated temperature cycling. This paper investigates the evolution of properties and damage mechanisms of 2D and 2.5D SiC/SiC composites after different thermal shock cycles at 1300 °C. It was found that SiC/SiC composites exhibit a temporary weight increase at the beginning of the thermal shock cycle. Subsequently, the weight decreases and stabilizes as the number of thermal shocks increases. After 10–30 thermal shocks at 1300 °C, the density of SiC/SiC composites decreases while the porosity increases. The fracture behavior of SiC/SiC composites remains the same as that of the initial composites, with 2D SiC/SiC composites exhibiting pseudo-tough fracture and 2.5D SiC/SiC composites exhibiting brittle fracture. The residual strength of the SiC/SiC composites decreased slightly while the residual modulus remained stable. This is related to the reaction of the SiC, and BN with water and oxygen at high-temperature. The thermal shock damage of SiC/SiC composites is anisotropic and is influenced by the structure of the fiber preforms. This is due to the difference in thermal expansion coefficients of the components within the composites and the structural properties of the fiber preforms.

1. Introduction

SiC/SiC composites are highly promising for aerospace applications as high-temperature components of aircraft engines, such as guide vanes, combustors, and tail nozzles [1,2]. These components are subjected to sudden temperature changes during operation, which can generate thermal stresses exceeding the yield stress of the matrix, resulting in thermomechanical damage to the composite [3,4]. The ability of a material to withstand sudden temperature changes is referred to as thermal shock resistance (TSR). It is necessary to investigate the thermal shock resistance of SiC/SiC composites, which will contribute to the life assessment and safety and reliability evaluation of SiC/SiC composite components.

Thermal shock tests are usually carried out in high-temperature furnaces, followed by water-quenching experiments [5–10]. Wang [11] et al. investigated the variation of mechanical properties of SiC/SiC composites containing different matrices and different fibers at different temperature differentials and analyzed the thermal shock damage

mechanism of the composites. Yutaka Kagawa [12] et al. investigated the thermal shock behavior of 2D SiC/SiC composites treated at 800 °C. Sirajus [13] et al. investigated the thermal shock behavior of 2D SiC/SiBN (PIP) composites at 800 °C and supported by acoustic emission techniques to analyze the thermal shock damage of the composites. Wu [14] et al. investigated the thermal shock behavior of 3D CVI SiC/SiC composites at 1200 °C. When the number of cycles exceeded 50 cycles, the microcracks in the composite reached saturation and the material properties did not degrade. Yang [15] et al. compared the thermal shock behavior of 2D and 3D SiC/SiC composites, the thermal shock damage characteristics of SiC/SiC composites were directly influenced by the woven structure of the fiber preforms.

It can be seen that different types of interfaces, SiC fibers and SiC matrix, and types of fiber preforms have different effects on the thermal shock resistance (TSR) of SiC/SiC composites. However, most of the studies on the thermal shock resistance of SiC/SiC composites focus on the mid-temperature range (800–1200 °C), and there is a lack of studies on the property evolution and failure mechanism of SiC/SiC composites

* Corresponding author.

** Corresponding author.

E-mail addresses: yu_heart@xmu.edu.cn (Y. Yu), hly@xmu.edu.cn (L. Huang).

<https://doi.org/10.1016/j.ceramint.2024.06.263>

Received 11 May 2024; Received in revised form 3 June 2024; Accepted 18 June 2024

Available online 19 June 2024

0272-8842/© 2024 Elsevier Ltd and Techna Group S.r.l. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

under thermal shock at temperatures above 1200 °C.

A combined characterization of the properties of SiC/SiC composites in response to thermal shock can be used to evaluate their thermal shock resistance behaviors. In this paper, the properties and damage behavior of 2D and 2.5D SiC/SiC composites prepared by the CVI process have been investigated under different thermal shock cycles at 1300 °C. The effects of different thermal shock cycles on weight loss, density, porosity and flexural properties were analyzed. The SEM was used to observe the microstructure and the bending fracture morphology in different regions, while the EDS was used to observe the elemental content. The aim was to provide a microscopic analysis of the effects of thermal shock on the damage behavior of SiC/SiC composites and to provide basic data for optimizing the subsequent properties of SiC/SiC composites and their practical applications.

2. Experimental

2.1. Materials

Firstly, 2D and 2.5D SiC fiber woven fabrics were prepared from the third-generation SiC fibers, and the warp and weft densities of the 2.5D fiber preforms were 8 × 3 (roots/cm). The structures of the 2D and 2.5D fiber fabrics are shown in Fig. 1. The fiber fabrics is in the form of a flat plate. Its dimensions are 150 mm × 250 mm × 3 mm. The third generation SiC fiber is the near stoichiometric ratio Cansas 3303 SiC fiber and its properties are shown in Table 1. The chemical vapor infiltration (CVI) method was used to deposit the boron nitride (BN) interlayer on the fiber fabrics, using boron trichloride (BCl₃) and ammonia (NH₃) as precursors at a deposition temperature of 800 °C. The SiC matrix was deposited on the surface of the fiber woven fabric after preparation of the BN interlayer by the CVI process, using methyl trichlorosilane (MTS) as precursor and hydrogen as a carrier gas, at a deposition temperature of 1050 °C until the weight gain was <5%. According to the experimental requirements, 50 mm × 4 mm × 3 mm bending test specimens were obtained by cutting the sheets. SiC coatings were then deposited on the surface of the specimens using the CVI deposition furnace. Table 2 shows the physical parameters of the as-prepared 2D and 2.5D SiC/SiC composites.

2.2. Thermal shock tests

Specimens were heated to a specified temperature and then immersed in water for the thermal shock test. Specimens were placed in a furnace preheated to 1300 °C for 10 min to ensure homogenization and then quenched in water at a temperature of 20 °C. A thermal shock cycle is formed by each heating and shocking process. After each thermal shock cycle, the samples were inspected for visible damage such as delamination and chipping after cooling to room temperature. The thermal shock cycles were repeated for 10, 20 and 30 cycles.

Table 1

General properties of Cansas 3303 SiC fibers.

SiC fiber Unit	Density g/cm ³	Diameter μm	Tensile strength GPa	Young's modulus GPa
Cansas-3303	2.8	12	3.0	310

Table 2

Physical parameters of 2D and 2.5D SiC/SiC composites.

Physical parameters	Unit	2D SiC/SiC	2.5D SiC/SiC
Porosity	%	11.3	8.3
Bulk density	g/cm ³	2.58	2.69
Fiber volume fraction	%	36	42
Matrix volume fraction	%	52.7	49.7
Thickness of the BN interface	nm	400	400

2.3. Properties tests

To quantify the effects of thermal shock cycles on the mechanical properties of SiC/SiC composites, the weight loss, density, porosity, and bending properties of the composites in the fiber direction (weft direction) were investigated. The mass variation of SiC/SiC composites was measured using a density electronic balance (BS210S, Beijing Sartorius Balance) after every 5 thermal shocks. The density and porosity of SiC/SiC composites were measured using the Archimedes method with a density electronic balance (BS210S, Sartorius Balance, Beijing) after every 10 thermal shocks. The bending strength σ and modulus E were tested at room temperature using an electronic versatile testing machine with a crosshead speed of 0.5 mm/min and a support span of 30 mm. The mid-span displacement of the beam was measured using a displacement sensor. Each test was performed on five specimens, and the final result was the average of the five specimens.

2.4. Characterization

Fracture surface analysis of the bend fractured specimens was performed in a scanning electron microscope (SEM, SU-70, Hitachi Tokyo Co., Ltd.) to examine topographical features and failure mode(s), and the composition was analyzed by an energy dispersive spectroscopy (EDS, Ultim Extreme, OXFORD, UK) attached to SEM.

3. Results and discussion

3.1. Microstructure and EDS results of the SiC/SiC composites after thermal shock

Fig. 2 shows the macroscopic morphology of the bending specimens of the 2D and 2.5D SiC/SiC composite after different thermal shock cycles. It can be seen that the surface of the SiC/SiC composites shows an obvious ablation phenomenon after thermal shock. The color of the surface of the thermally shocked specimens is significantly darker than

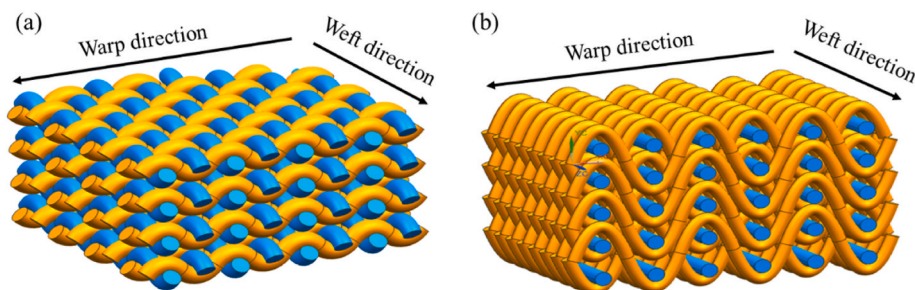


Fig. 1. The structures of (a) 2D and (b) 2.5D fiber fabrics.

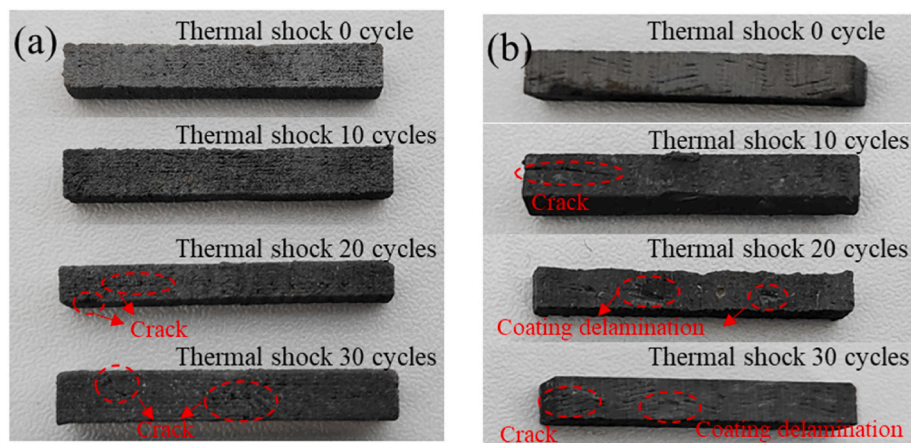


Fig. 2. Macroscopic morphology of the bending specimens of (a) 2D SiC/SiC composites and (b) 2.5D SiC/SiC composites after different thermal shock cycles.

that of the initial specimens. This is because the SiC coating on the surface was oxidized to form SiO_2 during thermal shock at 1300°C , which was further oxidized by water during water quenching. It can also be seen that the initial porosity of the SiC/SiC composites due to the preparation and processing is significantly reduced during the thermal shock because the SiO_2 produced by the continuous oxidation of the SiC coating during the thermal shock fills the original residual porosity. As the number of thermal shock cycles increases, two typical types of damage occur on the surface of SiC/SiC composites: cracking along the fiber orientation and coating delamination. This is due to the mismatch between the thermal expansion coefficients of the composite components and the oxidation of the SiC coating.

The thermal shock resistance of SiC/SiC composites is determined by the performance of the coating as the outermost barrier of the SiC/SiC composites. Fig. 3 shows the cross-sectional SEM morphology and EDS results of the SiC coating after 30 thermal shock cycles. It can be seen that the boundary between the SiC coating and the SiC fibers is visible and the thickness of the SiC coating is approximately $200\ \mu\text{m}$. In the coating region near the sample surface, the contact with water increased, leading to strong oxidation and corrosion reactions, and the EDS spectra after the thermal shock cycles also show that a large amount of SiC on the surface of the coating is oxidized to SiO_2 . As the number of thermal shock cycles increases, microcracks begin to appear on the coating, allowing oxygen to be released. As the number of thermal shock cycles increased, microcracks began to appear in the coating, allowing oxygen to enter the composite.

As a result, microcracks in the coating may allow oxygen to enter the interior of the SiC/SiC composite. The total time for each thermal shock is very short, and the exposure time of the composites to high temperatures and contact with water may not be sufficient to subject all of the

SiC and BN to damage by oxidation and water quenching. Therefore, both non-oxidized and oxidized morphology of fibers and interface can be observed using SEM.

First, the morphology of the unoxidized fibers and interface inside the SiC/SiC composite after thermal shock is shown in Fig. 4. It can be seen that the unoxidized fiber and interface morphology (Fig. 4 (a)–(c)) shows that the surface of the fiber is relatively smooth, and the BN interfacial layer and SiC matrix layer encapsulated on the surface of the fiber can be seen. However, as the number of thermal shocks increases, the SiC matrix on the surface of the clad fiber is oxidized into a SiO_2 layer to protect the BN interface inside from oxidation (Fig. 4 (b)–(c)). As the number of thermal shocks increases, the exposed SiC fibers begin to suffer oxidative damage (Fig. 4 (c)).

Secondly, Fig. 5 shows the morphology of fibers and interfaces oxidized in SiC/SiC composites after thermal shock. The surface of the fibers and BN interface of the non-shocked sample is relatively smooth (Fig. 5 (a)) and the boundary line between SiC, BN interface, and SiC matrix can be seen (Fig. 5 (e)). After 10 thermal shocks, the BN interface and SiC begin to oxidize and the thickness of the BN interface decreases. As the number of thermal shocks increases, the BN interface is consumed. The void created by BN depletion is filled by borosilicate glass and the SiC fibers are bonded to the SiC matrix by the borosilicate glass, enhancing the interfacial bond [16].

The oxidative damage of the fibers and interfaces was further analyzed using EDS and the EDS results after different thermal shock cycles for 2D and 2.5D SiC/SiC composites are shown in Figs. 6 and 7. It can be seen that Si, C, B, N, O are present on the fiber surface. The O element is mainly enriched on the fiber surface, which is consistent with the distribution pattern of the N element. This is due to the oxidation of BN on the fiber surface together with SiC fibers during thermal shock to

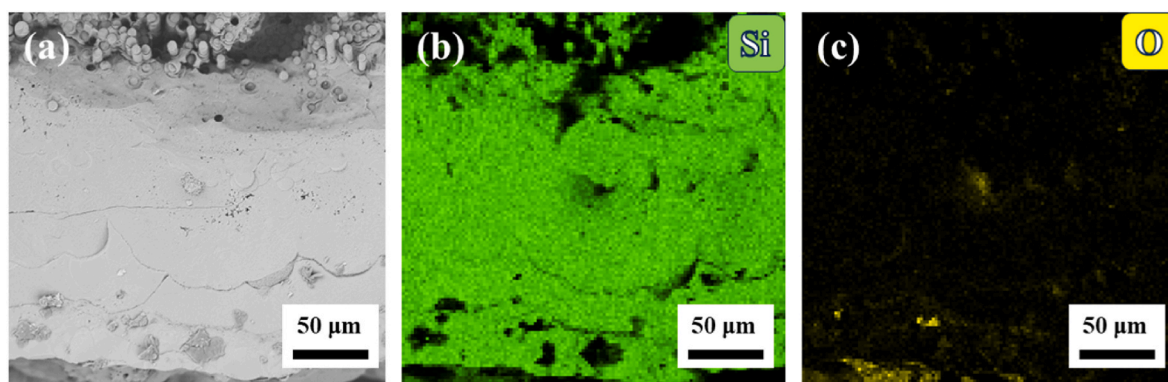


Fig. 3. The cross-sectional SEM morphology and EDS results of the SiC coating after 30 thermal shock cycles.

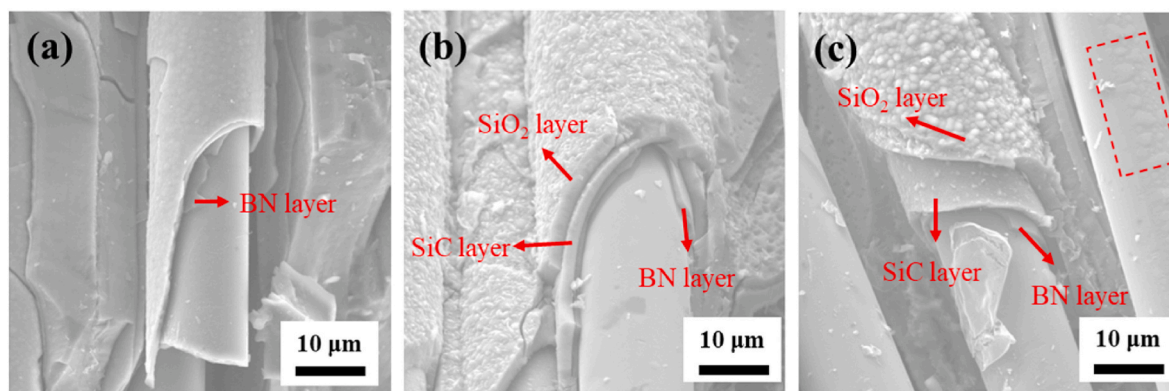


Fig. 4. Morphology of unoxidized fibers and interface inside SiC/SiC composites after thermal shock. (a) 10 cycles, (b) 20 cycles, (c) 30 cycles.

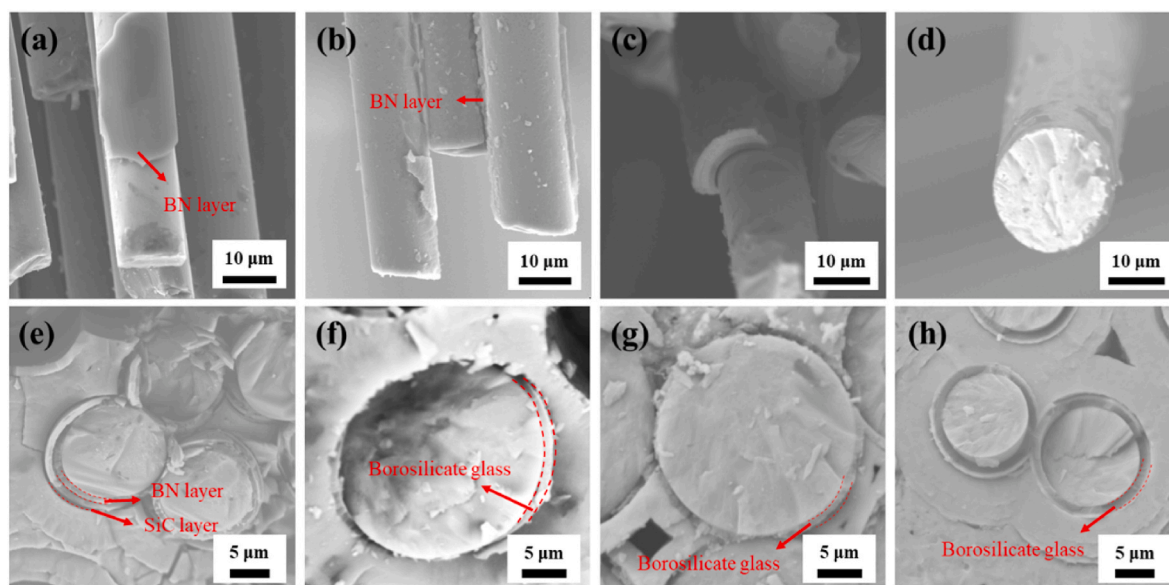


Fig. 5. Morphology of oxidized fibers and interfaces within SiC/SiC composites after thermal shock. (a) 2D, 0 cycles, (b) 2D, 10 cycles, (c) 2D, 20 cycles, (d) 2D, 30 cycles, (e) 2.5D, 0 cycles, (f) 2.5D, 10 cycles, (g) 2.5D, 20 cycles, (h) 2.5D, 30 cycles.

produce borosilicate glass.

3.2. Properties of the SiC/SiC composites after thermal shock

Fig. 8 presents the variation of weight loss of the samples with thermal shock cycles during thermal shocking of 2D and 2.5D SiC/SiC composites. It can be seen that the weight loss ratio of 2D and 2.5D SiC/SiC composites after 30 cycles of thermal shock is 1.18% and 0.88%, respectively. The weight loss ratio of 2.5D SiC/SiC composites is slightly lower than that of 2D SiC/SiC composites. The 2D and 2.5D SiC/SiC composites exhibit a small weight increase at the beginning of the thermal shock. This is caused by the oxidation of the coating. The weight loss curves of the SiC/SiC composites fluctuate slightly with the increasing number of thermal shocks after 5 thermal shock cycles, but the overall trend is increasing or stabilizing. This is due to the combination of coating delamination and oxidative consumption of each component.

Fig. 9 shows the variation of density and porosity with the thermal shock cycles during thermal shock of 2D and 2.5D SiC/SiC composites. It can be observed that the density of SiC/SiC composites shows a decreasing trend with the increase of thermal shock cycles, and the porosity shows an increasing trend with the increase of thermal shock cycles. During the thermal shock process of SiC/SiC composites, SiC is

continuously oxidized to form SiO_2 , and the density of SiO_2 is lower than that of SiC, which decreases the density of SiC/SiC composites. At the same time, the composites will produce cracks, holes, and other damage during thermal shock, which will increase the porosity of SiC/SiC composites.

Fig. 10 and Table 3 show the variation of bending strength with the number of thermal shocks of the 2D and 2.5D SiC/SiC composites. Meanwhile, the initial bending properties of the SiC/SiC composites are also shown in Table 3 and Fig. 10, as well as the bending properties of SiC/SiC composites after oxidation at 1300 °C in the air for 5 h (equivalent to the cumulative duration of 30 times thermal shock at 1300 °C). It can be seen that the bending strength of the 2D and 2.5D SiC/SiC composites showed a continuous decrease with an increasing number of thermal shocks. The bending strength of 2D SiC/SiC composites decreased from 526 MPa before thermal shock to 498 MPa (10 cycles), 473 MPa (20 cycles) and 463 MPa (30 cycles). The bending strength of 2.5D SiC/SiC composites decreased from 643 MPa before thermal shock to 589 MPa (10 cycles), 568 MPa (20 cycles) and 555 MPa (30 cycles). In addition, the bending strength of the composites after oxidation at 1300 °C for 5 h is slightly lower than the initial bending strength and higher than the strength after 30 thermal shocks, indicating that the oxidative damage to the composites after 30 thermal shocks is higher than that after oxidation for 5 h. When the SiC/SiC composites

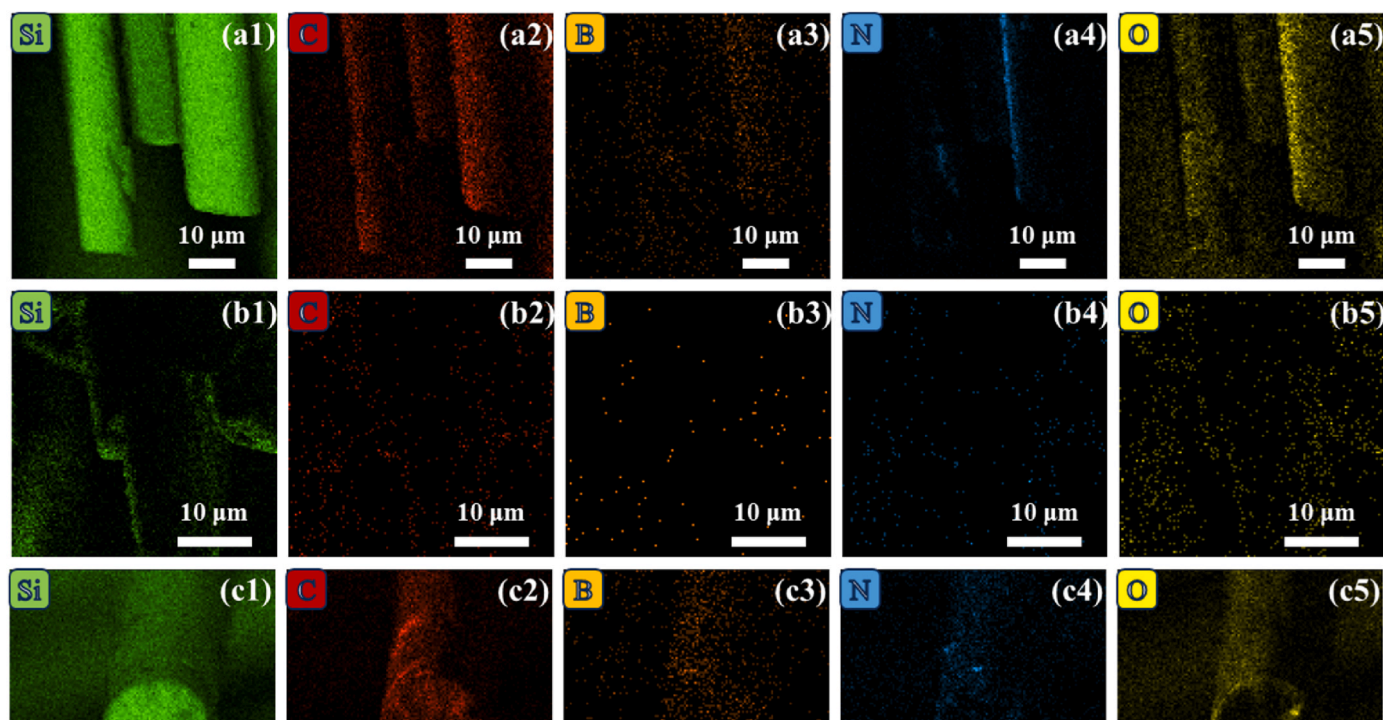


Fig. 6. The EDS consequences of fibers and interfaces of 2D SiC/SiC composites after thermal shock (a) 10 cycles, (b) 20 cycles, (c) 30 cycles.

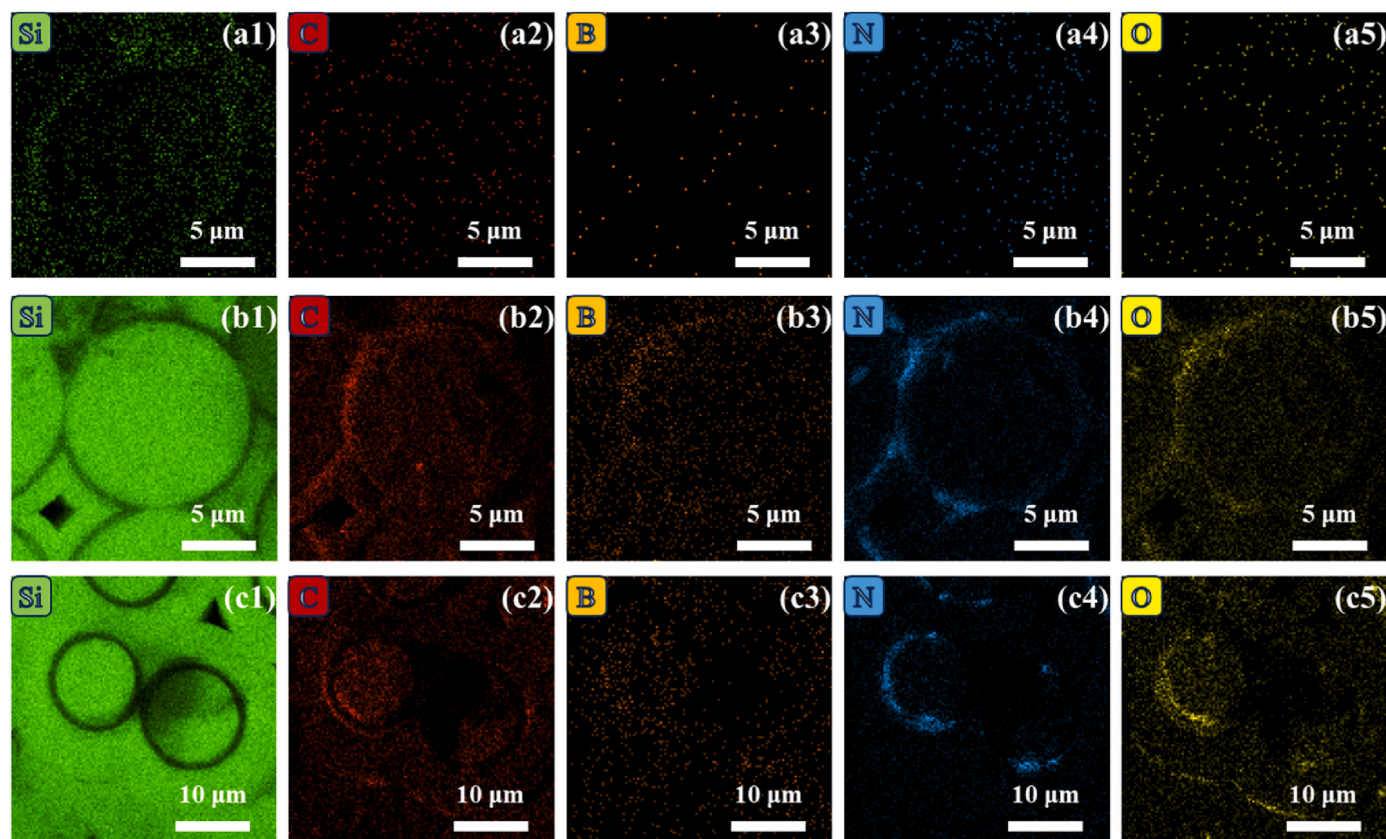


Fig. 7. The EDS consequences of fibers and interfaces of 2.5D SiC/SiC composites after thermal shock (a) 10 cycles, (b) 20 cycles, (c) 30 cycles.

are exposed to water during thermal shock, the water contacting the sample surface immediately evaporates to water vapor. Therefore, although the time required for 30 thermal shocks is approximately the

same as the time required for 5 h of oxidation, for SiC/SiC composites the reaction of the composite material with water vapor must be considered in addition to the reaction with oxygen during thermal

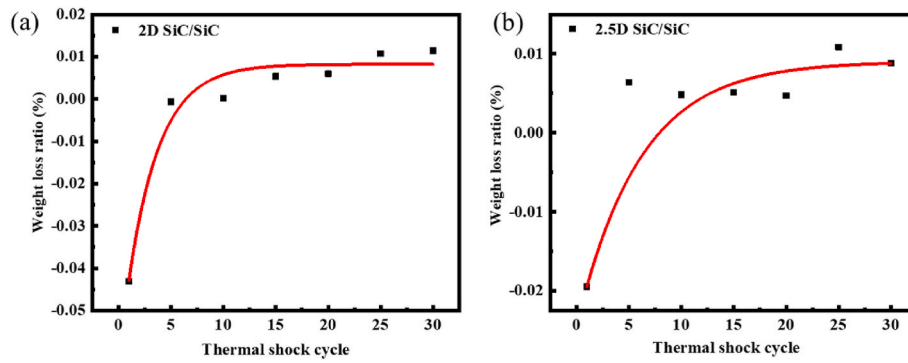


Fig. 8. The weight-loss curves of SiC/SiC composites during thermal shock tests. (a) 2D SiC/SiC, (b) 2.5D SiC/SiC.

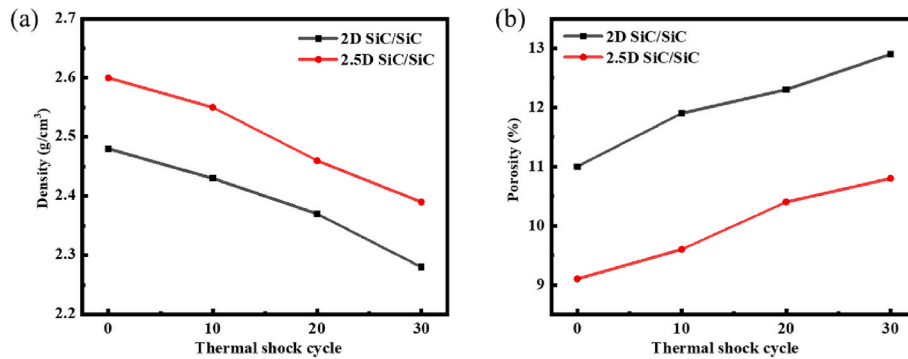


Fig. 9. The variations of density and porosity of 2D and 2.5D SiC/SiC composites during thermal shock tests. (a) Density, (b) Porosity.

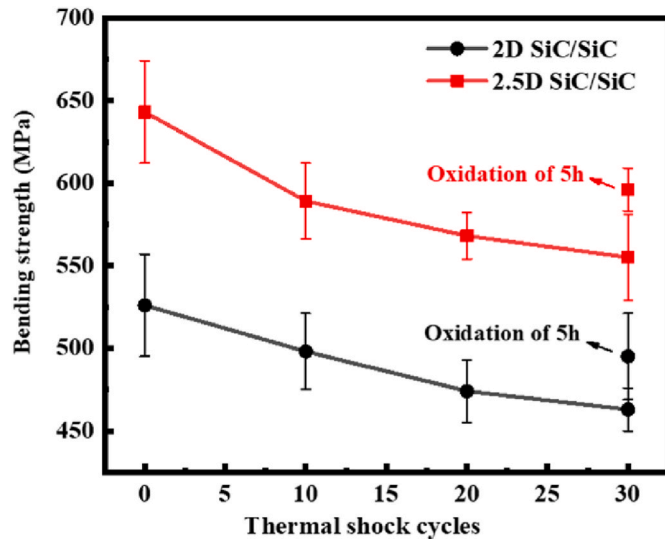


Fig. 10. The variation of bending strength with the number of thermal shocks of the 2D and 2.5D SiC/SiC composites.

Table 3

Bending strength of SiC/SiC composites with thermal shock cycles.

Heat treatment conditions	2D SiC/SiC	2.5D SiC/SiC
Thermal shock for 0 cycle(MPa)	526 ± 31	643 ± 31
Thermal shock for 10 cycles(MPa)	498 ± 23	589 ± 23
Thermal shock for 20 cycles(MPa)	473 ± 19	568 ± 14
Thermal shock for 30 cycles(MPa)	463 ± 13	555 ± 26
Thermal treated for 5 h(MPa)	495 ± 26	596 ± 13

shock, whereas for oxidation only the reaction with oxygen must be considered.

Fig. 11 shows the bending stress-strain curves of 2D and 2.5D SiC/SiC composites with different thermal shock cycles. As can be seen, the stress-strain curves of the SiC/SiC composites before and after thermal shock are not significantly different. There is a clear crack extension process of the 2D SiC/SiC composite before fracture, and a stair-step decrease in the curve due to fiber breakage within the composite accompanies the loading process. This is due to the fact that the reinforcement fibers of 2D SiC/SiC composites are mainly arranged in an in-plane (X-Y plane) with no fibers in the thickness direction and no fiber laps between layers. The uniform distribution of fibers in the in-plane helps the cracks to deflect and bifurcate into multiple paths, absorbing more energy and leading to ductile fracture. The stress-strain curve of 2.5D SiC/SiC composites has a distinct linear phase from loading to fracture. When the load reaches its maximum value, the curve drops suddenly. This is due to the complex structure of the 2.5D SiC/SiC composites, which tends to introduce internal stresses into the composites during the fiber preform and CVI processes, leading to easier crack initiation and propagation under flexural loading. Subsequently, the composites continue to be loaded and the curve gradually decreases, at which point the 2.5D SiC/SiC composites are still ductile. In general, the change in modulus is more representative of the overall damage state of the material than the strength, because the strength is related to the critical crack length, whereas the change in modulus reflects the number of cracks generated within the material [14]. It can also be seen from Fig. 11 that the modulus of 2D and 2.5D SiC/SiC composites changes very little before and after thermal shock, and the maximum strain at fracture of SiC/SiC composites after thermal shock is significantly lower after thermal shock. It can be assumed that the damage to the composites caused by the thermal shock is on the surface of the specimens and that the thermal shock has less effect on the damage inside the composites. In addition, the crack extension process on the

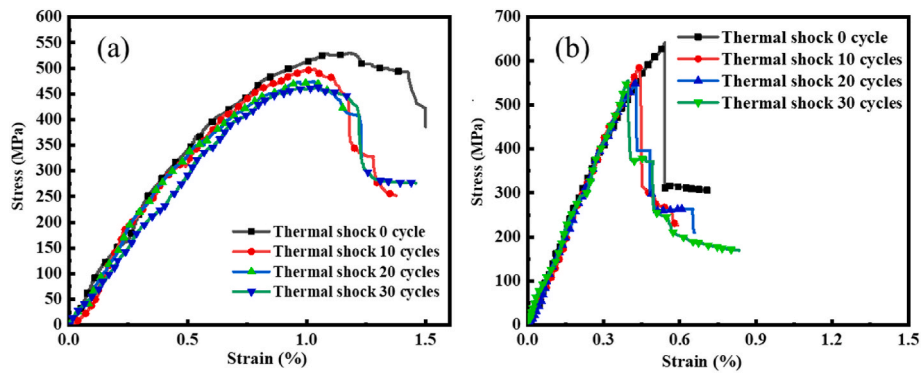


Fig. 11. The bending stress-strain curves of SiC/SiC composites after thermal shock tests at 1300 °C(a) 2D SiC/SiC, (b) 2.5D SiC/SiC.

stress-strain curve of 2D SiC/SiC composites after thermal shock is significantly shortened, and the fracture behavior of 2D SiC/SiC composites is gradually excessive from pseudo-toughening fracture to brittle fracture. This is due to the fact that as the number of thermal shock cycles increases, the BN interface is consumed, the bond between the fibers and the matrix is strengthened and the fibers are less likely to be pulled out. Thermal shock can affect the toughening mechanism of SiC/SiC composites.

Fig. 12 shows the fracture SEM morphology of 2D and 2.5D SiC/SiC composites after thermal shock cycles. It is observed that a large number of fibers are extracted from the 2D SiC/SiC composite, which is typical of pseudo-toughness fracture. The surface of the extracted fibers of the samples after thermal shock is covered with SiC matrix fragments and the matrix is damaged. 2.5D SiC/SiC composites without thermal shock bending fracture, there is a small amount of fiber pull-out phenomenon, and the pull-out length is short. As the number of thermal shocks increases, the length and number of fiber pull-out decreases, and the bending area is relatively flat, which is a typical brittle fracture.

3.3. Influence of fiber preform structures on thermal shock performance

Fig. 13 shows the residual strength of 2D and 2.5D SiC/SiC composites as a function of thermal shock cycles. It can be observed that the bending strength retention of 2D SiC/SiC composites after 10, 20, and 30

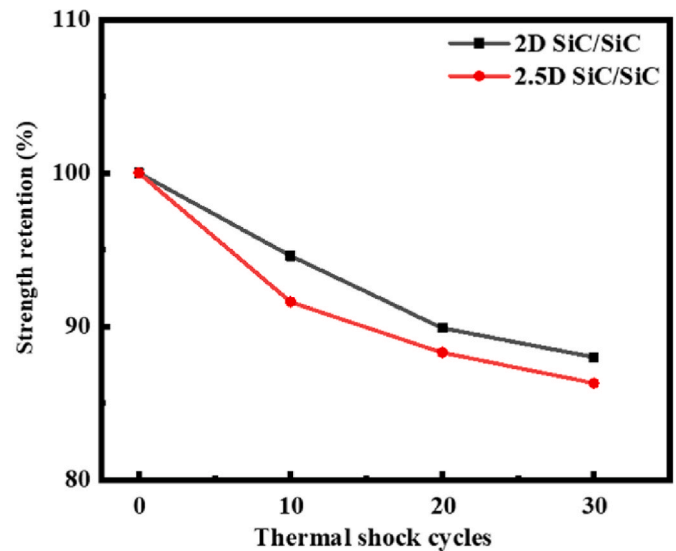


Fig. 13. Relation between thermal shock cycles and residual strength of 2D and 2.5D SiC/SiC.

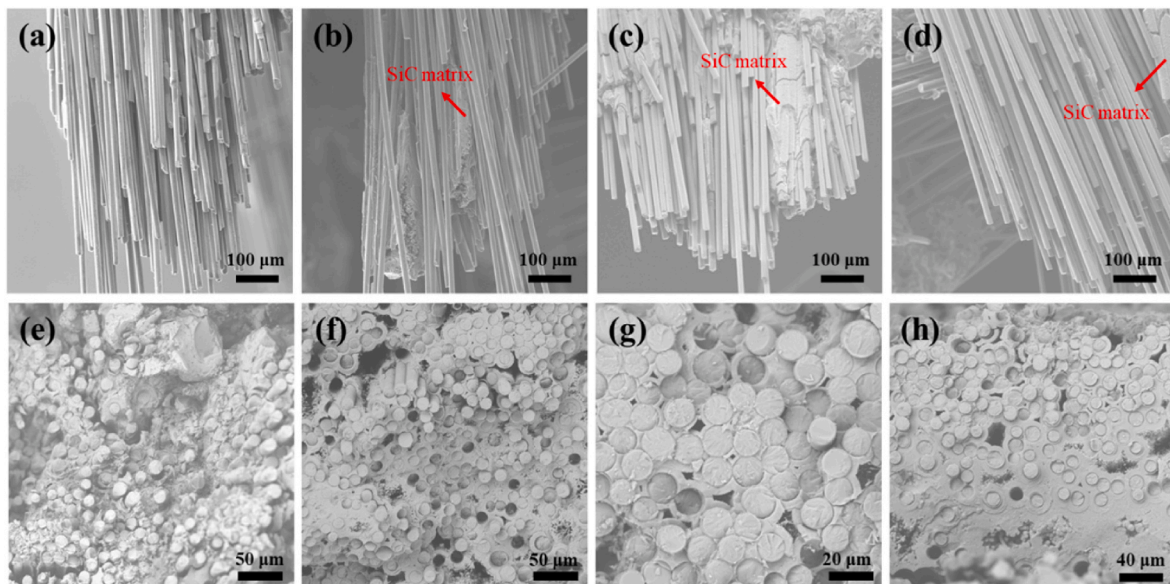


Fig. 12. The fracture SEM morphology of 2D and 2.5D SiC/SiC composites after thermal shock cycles (a) 2D, 0 cycles, (b) 2D, 10 cycles, (c) 2D, 20 cycles, (d) 2D, 30 cycles, (e) 2.5D, 0 cycles, (f) 2.5D, 10 cycles, (g) 2.5D, 20 cycles, (h) 2.5D, 30 cycles.

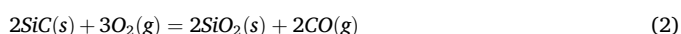
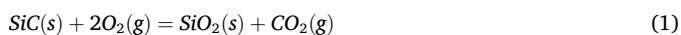
thermal shock cycles were 94.6%, 89.9%, and 88%, respectively. While the bending strength retention of 2.5D SiC/SiC composites after 10, 20, and 30 cycles of thermal shock were 91.6%, 88.3%, and 86.3%, respectively. The strength degradation of 2.5D SiC/SiC composites is greater than that of 2D SiC/SiC composites after suffering from thermal shock. This indicates that the degree of thermal shock damage of 2.5D SiC/SiC composites is greater than that of 2D SiC/SiC composites. It can also be seen from the macroscopic photographs of the bending specimens before and after thermal shock (Fig. 2) that more damage occurred on the surface of the 2.5D SiC/SiC composites compared to the 2D SiC/SiC composites, such as cracking along the direction of the fibers and peeling of the coating. It can also be further confirmed that the thermal shock damage of 2.5D SiC/SiC composites is greater than that of 2D SiC/SiC composites.

For SiC/SiC composites, when the temperature difference reaches a critical value (ΔT_c), the difference in the coefficient of thermal expansion (CTE) between the fibers and the matrix is directly related to the thermal stress that cracks the matrix [14]. The CTE of the CVI-SiC matrix is $4.6 \times 10^{-6}/^{\circ}\text{C}$ [17], while the CTE of the Canas 3303 SiC fibers is slightly lower than that of the CVI-SiC matrix. On the one hand, during water quenching of the SiC/SiC composites, the SiC matrix shrinks more than the SiC fibers, resulting in tensile stresses in the SiC matrix. As a result, a large number of microcracks perpendicular to the fiber direction are generated in the SiC matrix, and as the number of thermal shock cycles increases, the microcracks continue to expand to saturation. On the other hand, when SiC/SiC composites are heated, the SiC matrix is subjected to compressive stress, and the SiC fibers are subjected to tensile stress. When the tensile stress exceeds the strength limit of the fibers, the fibers break. During the subsequent thermal shock process the SiC matrix continues to crack until it fails.

The above results show that the thermal expansion coefficient has a significant effect on the thermal shock behavior of the SiC/SiC composites. Anisotropy of thermal shock damage is observed for 2D and 2.5D SiC/SiC composites. For 2D SiC/SiC composites, the thermal expansion coefficients of the composites in warp and weft directions are the same. For 2.5D SiC/SiC composites, the thermal expansion coefficients of the composites in the warp direction are larger than those in the weft direction, and the mechanical properties of 2.5D SiC/SiC composites in the warp direction are better than those in the weft direction. This can also be illustrated by the characteristics of the structures of the 2D and 2.5D fiber preforms shown in Fig. 1, where the higher fiber volume fraction in the warp direction determines the higher mechanical properties in that direction. Accordingly, during the thermal shock process, the 2.5D SiC/SiC composites generate more internal thermal stresses in the composites than the 2D SiC/SiC composites due to the mismatch between the thermal expansion coefficients and the structure characteristics of the fiber woven, which makes the thermal shock damage of the 2.5D SiC/SiC composites greater than that of the 2D SiC/SiC composites.

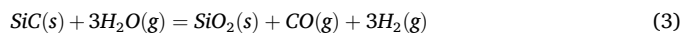
3.4. Possible mechanical property degradation mechanisms of composites in thermal shock

The thermal shock process of SiC/SiC composites includes the reaction of SiC/SiC composites with O_2 at high temperatures and the reaction with O_2 and H_2O during water quenching. The latter takes less time. At 1300°C in dry oxygen, the SiC coatings on the surface of SiC/SiC composites oxidized to form the SiO_2 , as per Eq. (1) [16,18]. SiO_2 surrounds the SiC matrix and fibers, which can slow down the oxidation reaction inside the composites.

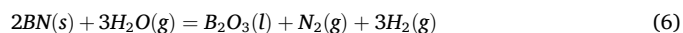
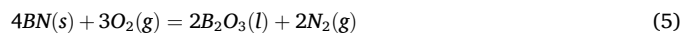


When the SiC/SiC composites were quenched with water, the SiC

coating and SiO_2 of the SiC/SiC composites reacted with water, as per Eq. (3) and Eq. (4). SiO_2 reacted with water to form $\text{Si}(\text{OH})_4$ gas.



As the SiC coating peeled or cracked, O_2 and H_2O would enter the interior of the composites and react with the components of the composite, such as the BN interface, the SiC fibers, and the SiC matrix. BN will react with O_2 and H_2O to form liquid B_2O_3 , as per Eq. (5) and Eq. (6).

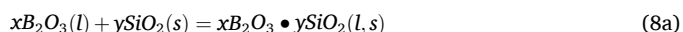


Three reactions take place in liquid B_2O_3 during thermal shock.

i). Liquid B_2O_3 will volatilize at elevated temperatures, as per Eq. (7).



ii). B_2O_3 reacts with SiO_2 to form borosilicate glass, given by Eq. (8).



iii). B_2O_3 undergoes various reactions with water to produce volatile boron hydroxide species [19–21], given by Eqs. (9–11).



Above all, SiC/SiC composites need to consider the corrosion of SiC/SiC composites in dry and wet oxygen environments in addition to enduring the internal thermal stresses due to high and low temperature cycling during the thermal shock process. The addition of water exacerbates the oxidative damage to SiC/SiC composites [18,22]. During thermal shock testing of SiC/SiC composites, weight loss and weight gain occur. Weight loss is caused by the reaction of SiO_2 with H_2O to form $\text{Si}(\text{OH})_4$ and the reaction of B_2O_3 and H_2O to form volatile boron hydroxide species. Weight increase is caused by the reaction of SiC coatings, SiC substrates and SiC fibres with O_2 or H_2O to form SiO_2 , and the reaction of BN with O_2 or H_2O to form B_2O_3 .

From the above discussions, the degradation of current SiC_f/SiC composites in the thermal shock at 1300°C can be proposed as follows. It can be mainly divided into five stages, which are schematically shown in Fig. 14.

Stage I. Oxidation of SiC coating occurs. The SiC coating reacts with O_2 to form SiO_2 in an isothermal environment at 1300°C . During the isothermal process, the weight of SiO_2 increases and its density is lower than that of SiC, causing the density of SiC/SiC composites to decrease. There is a significant volume expansion and pore-filling effect as SiC oxidizes to SiO_2 , which can hinder oxygen penetration [23].

Stage II: Cracking of SiC coating after tensile stress. When SiC/SiC composites change rapidly from high temperature to low temperature, the composites will have a large temperature gradient from the center to the surface, resulting in tensile stress on the surface of the composites and compressive stress inside. The surface coating of SiC/SiC composites will be the first to crack under the action of tensile stress.

Stage III: Reaction of SiC coating and SiO_2 with H_2O and O_2 . SiC/SiC composites are removed from high temperature and quenched with water. Water and O_2 react with the SiC and SiO_2 on the surface

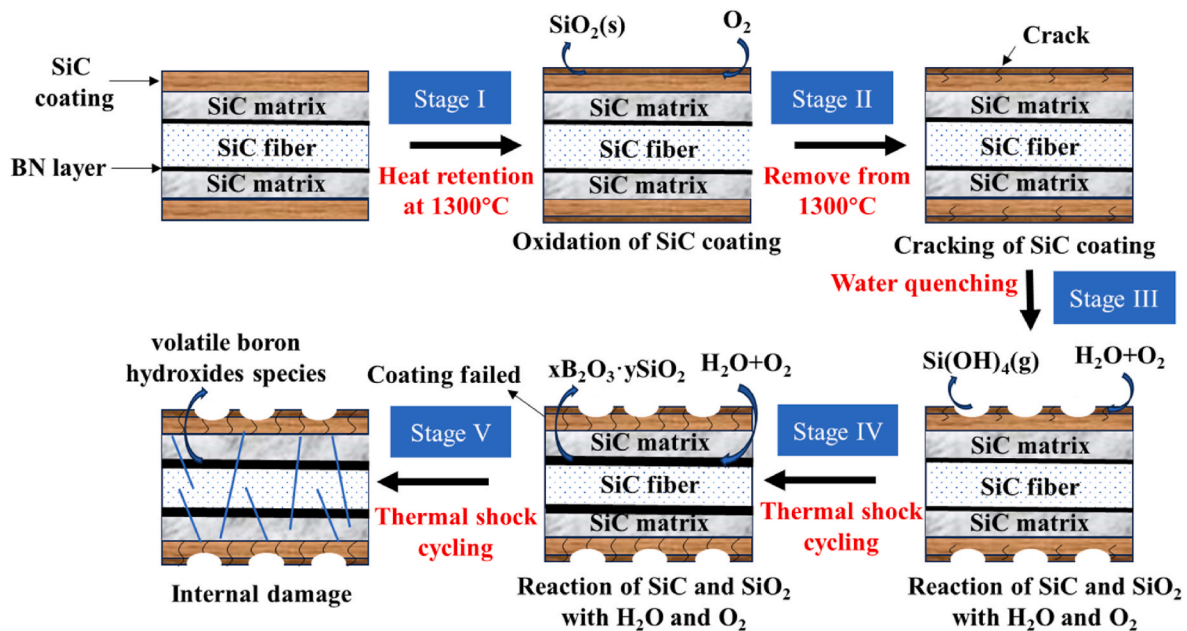


Fig. 14. A schematic drawing of the damage behavior and degradation mechanism of SiC/SiC composites under thermal shock at 1300 °C.

of the coating to form the gas Si(OH)_4 . The gas escapes and forms corrosion scores on the surface of the specimen as shown in Fig. 2.

Stage IV: The SiC coating fails and H_2O and O_2 enter the interior of the composite to react with the SiC fibers, SiC matrix, and BN interface. The composite repeats the damage of stage II and stage III after multiple thermal shocks, resulting in failure of the SiC coating. External gases (e.g., H_2O and O_2) can enter the composite through the holes and react with the SiC fibers, SiC matrix, and BN interface. The reaction produces $\text{B}_2\text{O}_3(\text{s})$, SiO_2 , and their oxidation product borosilicate glass, which fills the holes left by the escaping gases, allowing the SiC fibers to form a strong bond with the SiC matrix.

Stage V: Internal stresses cause damage to the composites. Volume expansion of oxidation products and the internal stresses from high and low-temperature cycling will increase the internal stresses in the composites after multiple thermal shocks, exacerbating the internal damage in the composites.

3.5. The implication

Considering the practical service environment of SiC/SiC composites in aircraft engines, high and low temperature cycling is inevitable [23]. The transition from $\alpha\text{-SiO}_2$ to $\beta\text{-SiO}_2$ (low to high temperature) and from $\beta\text{-SiO}_2$ to $\alpha\text{-SiO}_2$ (high to low temperature) accompanies the high and low temperature cycling process [23]. The resulting volumetric expansion and contraction of the internal components of the composites resulted in large residual internal stresses within the composites, which exacerbated the damage to the composites. Therefore, the first implication of this study is that SiC/SiC composites need to improve the thermal shock resistance properties of their coatings when operating in thermal shock environments. To improve the thermal shock resistance of SiC/SiC composites, research on thermal barrier coatings has been a major focus in recent years [24–26]. Improving the performance of the outer protective coating can prevent the generation of excessive SiO_2 on the surface of the coating or prevent the external corrosive gases from entering the interior of the SiC/SiC composites. Secondly, H_2O and O_2 will enter into the interior of the composite if the outer protective coating fails. The BN interface and SiC undergo a series of reactions with H_2O and O_2 to produce gases such as CO , CO_2 , H_2 , Si(OH)_4 , $\text{B}_2\text{O}_3(\text{g})$, N_2 , and volatile boron hydroxide species, which leave the pores as the gases escape. The $\text{B}_2\text{O}_3(\text{s})$ and SiO_2 and their oxidation products borosilicate

glass, will fill these pores and form a strong bond with the SiC fibers. An increase in the number of thermal shock cycles results in the thickening of the borosilicate glass, volume expansion, and an increase in the internal stress of the SiC/SiC composites, which tends to produce defects such as cracks within the SiC/SiC composites. Based on this, the second implication of this study is to avoid direct contact between SiC fibers and borosilicate glass or to avoid the formation of a strong bond between the two, which can be improved by doping and other strategies to increase the stability of borosilicate glass. It is also necessary to reduce the difference in the thermal expansion coefficient of the individual components of the SiC/SiC composites to reduce the residual internal stresses caused by thermal shock during thermal shock and thus reduce the damage to the composites.

4. Conclusion

In this work, the damage behavior and corresponding mechanisms of 2D and 2.5D SiC/SiC composites prepared by the CVI process under thermal shock at 1300 °C have been studied. The main conclusions are as follows.

- (1) The SiC/SiC composites are accompanied by weight loss and weight increase variations during thermal shock and show an overall trend of weight loss. The weight increase was mainly caused by the formation of the SiO_2 . The weight loss included gas emission, coating peeling and other phenomena.
- (2) During thermal shock, the SiC/SiC composites continuously produce the gas to escape and generate SiO_2 inside the composites. The density of the composites decreases as the number of thermal shock cycles increases, and the porosity increases as the number of thermal shock cycles increases.
- (3) The SiC/SiC composites exhibit the same flexural mechanical behavior and flexural modulus as the original composites. Thermal shock affects the mechanical properties of SiC/SiC composites, and the bending strength decreases with the increase of thermal shock cycles. In addition, the composites show anisotropy in the thermal shock damage, and the structure of the fiber preforms will affect the thermal shock resistance of SiC/SiC composites.

- (4) During the thermal shock process, the peeling off of the coating and the high-temperature reaction of the composite components with oxygen and water are the main reasons for the degradation of the properties of SiC/SiC composites.

Data availability

The data that has been used is confidential.

CRediT authorship contribution statement

Yongjin Wei: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis. **Junwei Lan:** Writing – review & editing, Investigation. **Yuxi Yu:** Supervision, Resources, Methodology, Conceptualization. **Liuying Huang:** Supervision, Resources, Methodology, Conceptualization. **Dahai Ye:** Resources. **Zhongyi Fu:** Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (51675452) and the Natural Science Foundation of Fujian Province of China, Grant/Award Number: No. 2022J01041.

References

- [1] H. Yu, X. Zhou, Z. Wei, et al., Mechanical behavior of SiC_f/SiC composites with alternating PyC/SiC multilayer interphases, *Mater. Des.* 44 (2013) 320–324, <https://doi.org/10.1016/j.matdes.2012.07.073>.
- [2] R. Naslain, Design, preparation and properties of non-oxide CMCs for application in engines and nuclear reactors: an overview, *Compos. Sci. Technol.* 64 (2004) 155–170, [https://doi.org/10.1016/S0266-3538\(03\)00230-6](https://doi.org/10.1016/S0266-3538(03)00230-6).
- [3] A. Parvanyan, H. Salimijazi, M. Fathi, et al., Synthesis and thermal shock evaluation of porous SiC ceramic foams for solar thermal applications, *J. Am. Ceram. Soc.* 102 (2018) 2009–2020, <https://doi.org/10.1111/jace.16007>.
- [4] M. Ruggles-wrenn, R. Lanser, Tension-compression fatigue of an oxide/oxide ceramic composite at elevated temperature, *Mater. Sci. Eng.* 659 (2016) 270–277, <https://doi.org/10.1016/j.msea.2016.02.057>.
- [5] H. Kou, W. Li, T. Cheng, et al., Thermal shock resistance of ultra-high temperature ceramics under active cooling condition including the effects of external constraints, *Appl. Therm. Eng.* 110 (2017) 1247–1254, <https://doi.org/10.1016/j.applthermaleng.2016.09.004>.
- [6] C. Zhang, H. Wang, Y. Liu, et al., Thermal shock resistance of a 2D-C/SiC composite and its damage mechanisms, *Adv. Appl. Ceram.* 112 (2014) 499–504, <https://doi.org/10.1179/1743676113Y.0000000120>.
- [7] C. Zhang, X. Wang, B. Wang, et al., Thermal shock properties of a 2D-C/SiC composite prepared by chemical vapor infiltration, *J. Mater. Eng. Perform.* 22 (2012) 1680–1687, <https://link.springer.com/article/10.1007/s11665-012-0465-z>.
- [8] X. Zeng, Y. Yang, Q. Deng, et al., Thermal shock response behaviour of carbon fiber reinforced silicon carbide matrix composites prepared by chemical vapour infiltration at different heating rates, *Ceram. Int.* 47 (2021) 31457–31469, <https://doi.org/10.1016/j.ceramint.2021.08.022>.
- [9] H. Mei, L. Cheng, Damage analysis of 2D C/SiC composites subjected to thermal cycling in oxidizing environments by mechanical and electrical characterization, *Mater. Lett.* 59 (2005) 3246–3251, <https://doi.org/10.1016/j.matlet.2005.05.052>.
- [10] H. Mei, L. Cheng, Thermal cycling response behavior of ceramic matrix composites under load and displacement constraints, *Mater. Sci. Eng.* 486 (2008) 235, <https://doi.org/10.1016/j.msea.2007.09.053>.
- [11] H. Wang, R. Singh, R. Lowden, Thermal shock behavior of two-dimensional woven fiber-reinforced ceramic composites, *J. Am. Ceram. Soc.* 79 (2010) 1783–1792, <https://doi.org/10.1111/j.1151-2916.1996.tb07996.x>.
- [12] Y. Kagawa, Thermal shock damage in a two-dimensional SiC/SiC composite reinforced with woven SiC fibers, *Compos. Sci. Technol.* 57 (1997) 607–611, [https://doi.org/10.1016/S0266-3538\(97\)00005-5](https://doi.org/10.1016/S0266-3538(97)00005-5).
- [13] S. Salekeen, J. Amoko, H. Mahfuz, et al., Mechanical property degradation of a Nicalon fiber reinforced SiNC ceramic matrix composite under thermal shock loading, *Compos. Struct.* 78 (2007) 477–485, <https://doi.org/10.1016/j.compstruct.2005.11.016>.
- [14] S. Wu, L. Cheng, L. Zhang, et al., Thermal shock behavior of a three-dimensional SiC/SiC composite, *Metall. Mater. Trans. A* 37 (2006) 3587–3592, <https://link.springer.com/article/10.1007/s11661-006-1053-3>.
- [15] J. Yang, E. Dittmars, W. Lin, et al., Thermomechanical durability of CVI-processed two-dimensionally woven and three-dimensionally braided SiC fiber-reinforced SiC composites, *J. Mater. Sci.* 29 (1994) 5491–5497, <https://link.springer.com/article/10.1007/BF01171567>.
- [16] K. Detwiler, E. Opila, Oxidation of SiC/BN/SiC ceramic matrix composites in dry and wet oxygen at intermediate temperatures, *J. Eur. Ceram. Soc.* 42 (2022) 4110–4120, <https://doi.org/10.1016/j.jeurceramsoc.2022.04.003>.
- [17] W. Yang, A. Kohyama, T. Noda, et al., Interfacial characterization of CVI-SiC/SiC composites, *J. Nucl. Mater.* 307–311 (2002) 1088–1092, [https://doi.org/10.1016/S0022-3115\(02\)00949-2](https://doi.org/10.1016/S0022-3115(02)00949-2).
- [18] L. Filipuzzi, G. Camus, R. Naslain, et al., Oxidation mechanisms and Kinetics of 1D-SiC/C/SiC composite materials: I, an experimental approach, *J. Am. Ceram. Soc.* 77 (2005) 459–466, <https://doi.org/10.1111/j.1151-2916.1994.tb07015.x>.
- [19] N. Jacobson, G. Morscher, D. Bryant, et al., High-temperature oxidation of boron nitride: II, boron nitride layers in composites, *J. Am. Ceram. Soc.* 82 (2004) 1473–1482, <https://doi.org/10.1111/j.1151-2916.1999.tb01944.x>.
- [20] N. Jacobson, S. Farmer, A. Moore, et al., High-temperature oxidation of boron nitride: I, monolithic boron nitride, *J. Am. Ceram. Soc.* 82 (2004) 393–398, <https://doi.org/10.1111/j.1151-2916.1999.tb0075.x>.
- [21] C. Cofer, J. Economy, Oxidative and hydrolytic stability of boron nitride - a new approach to improving the oxidation resistance of carbonaceous structures, *Carbon* 33 (1995) 389–395, [https://doi.org/10.1016/0008-6223\(94\)00163-T](https://doi.org/10.1016/0008-6223(94)00163-T).
- [22] E. Opila, Variation of the oxidation rate of silicon carbide with water-vapor pressure, *J. Am. Ceram. Soc.* 82 (2004) 625–636, <https://doi.org/10.1111/j.1151-2916.1999.tb01810.x>.
- [23] J. Zhang, R. Liu, Y. Jian, et al., Degradation mechanism of SiC_f/SiC composites after long-time water vapor and oxygen corrosion at 1300°C, *Corrosion Sci.* 197 (2022) 110099, <https://doi.org/10.1016/j.corsci.2022.110099>.
- [24] V. Leisner, K. Kelm, U. Schulz, Thin single-phase yttrium-based environmental barrier coating systems for SiC/SiC CMCs, *J. Eur. Ceram. Soc.* 42 (2022) 7275–7287, <https://doi.org/10.1016/j.jeurceramsoc.2022.08.035>.
- [25] J. Huang, R. Liu, Q. Hu, et al., High temperature abrasion sealing coating for SiCf/SiC ceramic matrix composites, *Ceram. Int.* 49 (2022) 1779–1790, <https://doi.org/10.1016/j.ceramint.2022.09.141>.
- [26] X. Yang, C. Feng, W. Qqing, ZrB₂-SiC as a protective coating for C/SiC composites: effect of high temperature oxidation on thermal shock property and protection mechanism, *J. Asian Ceram. Soc.* 4 (2016) 159–163, <https://doi.org/10.1016/j.jascr.2016.02.001>.